

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – DEBUGGING & OPTIMISATION TASK

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 3 – Debugging & Optimisation Task
Weightage:	10% of CIA	Total Marks:	100
CO2 Statement:	Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)		
Student Name:	M. Venkata Vasu Deva Reddy	Reg. No.:	192424182
Section / Batch:	AI & DS	Date:	

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given program/code segment before identifying errors.
- Clearly identify logical, syntactical, and performance-related issues in the code.
- Apply systematic debugging techniques to correct the errors effectively.
- Optimize the given solution by improving efficiency, reducing unnecessary computations, or enhancing time complexity wherever possible.
- Provide proper justification for all corrections and optimization decisions.
- Write clean, structured, and well-documented code with appropriate comments.
- Maintain clarity, logical reasoning, and technical accuracy throughout the solution.

Question Type	Focus Area	Questions
Debugging & Optimisation Task	Identify errors, debug programs, and optimize algorithm efficiency using appropriate techniques	Q1

SECTION A – DEBUGGING & OPTIMISATION TASK

Code of Conduct

I Vasu Deva Reddy (192424182)_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: vasu_____

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Identification	20%				
Debugging	25%				
Optimization	20%				
Analysis	20%				
Code Quality	15%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

This is 1st question answer:

Programiz

Python Online Compiler

Trade Crypto CFDs

Spreads from 0.0 pips

Leverage 1 to 1:500

Trade Now

Trading derivatives involves high risk to your capital.

main.py

Run

Share

Icons

```
21     return arr
22
23
24 # ----- Main Program -----
25
26 # Input from user
27 arr = list(map(int, input("Enter numbers separated by spaces: ")
28                       .split()))
29 print("Original Array:", arr)
30
31 # Sort the array
32 selection_sort(arr)
33
34 print("Sorted Array:", arr)
```

Output

Enter numbers separated by spaces: 45 34 23 56 76
Original Array: [45, 34, 23, 56, 76]
Sorted Array: [23, 34, 45, 56, 76]

=== Code Execution Successful ===

This is for 2nd question:

Programiz

C Online Compiler

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Trading derivatives involves high risk to your capital.

main.c

Run

Share

Icons

```
38     scanf("%d", &key);
39
40     int result = linearSearch(arr, n, key);
41
42     if (result != -1)
43     {
44         printf("Element found at index %d\n", result);
45     }
46     else
47     {
48         printf("Element not found in the array.\n");
49     }
50
51     return 0;
52 }
```

Output

Enter the number of elements: 4
Enter 4 elements:
29 39 49 59
Enter the element to search: 39
Element found at index 1

=== Code Execution Successful ===